

CS301 Paper Reviews Analysis

9 Student Reviews — Jun 2026 | Most Repeated Topics + Model Answers

9 Papers Analyzed

Most Repeated Topics

Model Answers Included

TOPIC FREQUENCY — KITNI BAAR AYA HAR TOPIC

Stack (Push/Pop/Output)

7 out of 9 papers

Almost every paper had stack-related output questions — push/pop sequence, top element, infix to postfix using stack.

Reference Variables & Pointers

6 out of 9 papers

int &y = x style reference variable output, pointer address printing, and functions returning references.

Queue (Enqueue/Dequeue/Operations)

6 out of 9 papers

Queue output, which operation does queue NOT support, dequeue and enqueue trace, MyQueue class output.

Infix to Postfix Conversion

5 out of 9 papers

Convert infix expression to postfix step-by-step using a table (Symbol | Postfix | Stack columns).

BST — Delete Node / Tree Diagram

4 out of 9 papers

Delete root node or any node from given BST and redraw the tree. Also: BST depth formula.

Arrays — Output / Code

3 out of 9 papers

Array loop output ($2*i+1$ pattern), array data types, removing element from linked list.

Linked List (Remove / Pointer Names)

3 out of 9 papers

Remove current node from linked list, pointer names in singly linked list, what fills the gap after removal.

ACTUAL QUESTIONS — PAPERS SE NIKALE HUE + MODEL ANSWERS

SHORT Reference Variable Output — `int &y = x`



Find the output of this program: (appeared in Reviews 01 and 02)

```
int main() {
    int x = 10;
    int &y = x;    // y is an alias for x
    y = y + 10;
    cout << x << endl << y;
    return 0;
}
```

Output:

20
20

Explanation: y is just another name for x. They share the same memory. When `y = y + 10` runs, it adds 10 to the shared memory location. So both x and y become 20.

Hint: *y = x ka alias hai — dono same memory mein hain. Ek badla, doosra automatically badal gaya.*

Review 01

Review 02

SHORT Pointer Address Output — &x vs ptr

Find the output: (Review 01)

```
int main() {
    int x = 5;
    int *ptr = &x;
    cout << &x << endl << ptr;
    return 0;
}
```

Output:

0x... (some memory address)
0x... (same memory address)

Both `&x` and `ptr` hold the same address — the memory location of variable `x`. The exact address will be something like `0x7ffee4b2c` — it changes every run but both lines will always print the same value.

Hint: *&x = x ka address. ptr = x ka address store karta hai. Dono same output denge — same address.*

Review 01

LONG Array Loop Output — 2*i+1 Pattern

Find the output: (Review 01 — note: the original code had a typo `i=0`, it should be `i=0`)

```
int main() {
    int x = 5;
    int array[x];
    for(int i = 0; i < x; ++i) {
        array[i] = 2*i + 1;
        cout << array[i] << endl;
    }
    return 0;
}
```

Output:

1
3
5
7
9

i goes from 0 to 4. Formula $2*i+1$ gives: $2(0)+1=1$, $2(1)+1=3$, $2(2)+1=5$, $2(3)+1=7$, $2(4)+1=9$.

Hint: $2*i+1$ formula = odd numbers ki sequence generate karta hai (1,3,5,7,9...). Yaad rakho.

Review 01

Review 05

LONG BST Depth Formula — n = 100,000 nodes ▼

(a) Write the formula to find depth d of a complete binary tree with n nodes.

Formula: $d = \log_2(n + 1) - 1$

(b) Find depth when n = 100,000

```
d = log2(100,000 + 1) - 1
d = log2(100,001) - 1
d = 16.609... - 1
d ≈ 16
```

Important discrepancy noted by a student: The handout page 121 shows $d \approx 20$, but the correct calculation using the formula gives $d \approx 16$. The student's calculation ($d \approx 16$) is mathematically correct. The handout value of 20 appears to be an error or uses a different approximation. Write $d \approx 16$ in your answer with the calculation shown.

Hint: Formula yaad karo: $d = \log_2(n+1) - 1$. Sirf yeh ek formula hai BST depth ke liye. Calculator allowed ho toh easy hai.

Review 01

SHORT Queue — Which Operation Does Queue NOT Support? ▼

MCQ / Short: Which of these is NOT a queue operation? Enqueue / Dequeue / Front / Pop

Answer: POP

Pop is a Stack operation, not a Queue operation. Queue uses Enqueue (add) and Dequeue (remove).

Queue supported operations: Enqueue, Dequeue, Front (peek), IsEmpty. Pop belongs to Stack.

Hint: Pop = Stack ka operation. Queue mein Dequeue hota hai. Yeh MCQ mein exact form mein aaya tha — yaad rakho.

Review 09

Review 04

SHORT MyQueue Class — Enqueue/Dequeue Output ▼

A MyQueue class enqueues 10, 20, 30 then dequeues once. What is the state of the queue? (Review 09)

```
MyQueue q;
q.enqueue(10);
q.enqueue(20);
q.enqueue(30);
q.dequeue();
// What is in the queue now?
```

Answer: Queue contains: 20, 30
Dequeue removes from the front (FIFO). 10 was added first, so 10 is removed. 20 and 30 remain.

Hint: Queue = FIFO. Pehle aaya pehle gaya. 10 pehle aaya tha, dequeue ne use hata diya.

Review 09 Review 02

LONG Infix to Postfix — Step by Step Table

Convert to postfix using the table method (Symbol | Stack | Postfix). Multiple papers had this exact format.
Example: $A^B * C + D$

How to fill the table: For each symbol, record (1) the symbol, (2) what the stack looks like after processing, (3) what has been written to postfix output so far.

Expression: $A^B * C + D$ (^ = exponent, highest precedence)

Symbol	Stack (top →)	Postfix Output
A	(empty)	A
^	^	A
B	^	AB
*	*	AB^
C	*	AB^C
+	+	AB^C*
D	+	AB^C*D
(end)	(empty)	AB^C*D+

Final Postfix: $AB^C * D +$

Also from papers: $6 + 4 * 3 / (6 - 2)$

Symbol	Stack	Postfix Output
6	(empty)	6
+	+	6
4	+	64
*	+ *	64
3	+ *	643
/	+ /	643*
(	+ / (	643*
6	+ / (	643*6
-	+ / (-	643*6

2	+ / (-	643*62
)	+ /	643*62 -
(end)	(empty)	643*62 - / +

Final Postfix: 643*62-/+

Hint: Table format: Symbol column mein ek ek character likho. Stack column mein current stack state. Postfix mein abhi tak ka output. Yeh exactly wahi format tha jo papers mein aaya.

Review 03

Review 06

Review 09

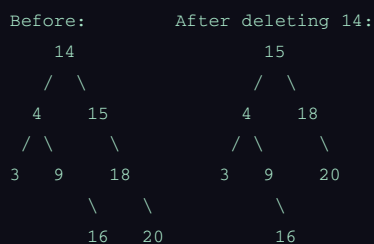
LONG BST — Delete Root Node and Redraw Tree

A BST is given. Delete the root node and redraw the tree. (Reviews 04, 06)

Algorithm for deleting a node with 2 children:

1. Find the inorder successor = the smallest node in the right subtree (go right once, then keep going left).
2. Copy the inorder successor's value into the node to be deleted.
3. Delete the inorder successor node (it will be a leaf or have 1 child — easier case).

Example — delete root (14) from this BST:



Steps:

1. Node 14 has 2 children — use inorder successor.
2. Right subtree of 14 starts at 15. Go left from 15 — no left child exists. So inorder successor = 15.
3. Copy 15 into root position. Delete original 15 node (it had only right child 18 — simple case).

Hint: Root delete karna = Case 3 (2 children). Inorder successor = right subtree ka leftmost node. Woh value root mein copy karo, phir use delete karo.

Review 04

Review 06

MCQ Linked List — Remove Current Node, What Fills Its Place?

MCQ: List is [12, 45, 17, 34, 51]. Current pointer is at 45. After removing 45, which number takes its place? (Review 03)

Answer: 17

When 45 is removed, the pointer from 12 now jumps directly to 17. So 17 effectively takes the position after 12. The list becomes [12, 17, 34, 51].

Hint: Remove = previous node ka next pointer, removed node ke next par point karta hai. 12 ka next 45 tha, ab 17 ho gaya.

Review 03

SHORT Write the pointer names used in a Singly Linked List

Short question from Review 06: Write the names of pointers used in a singly linked list.

Pointers in a Singly Linked List:

1. **head** — points to the first node of the list.
2. **current** (or curr) — points to the node currently being accessed/traversed.
3. **next** — inside each node, points to the next node.
4. **tail** (optional) — points to the last node for efficient insertion at end.

Hint: Singly linked list mein har node ka sirf ek next pointer hota hai. Head list ka entrance hai. Current traversal ke waqt use hota hai.

Review 06

LONG Function Returning Reference — Complex Output

Review 02 mentioned a question like this (student said exact code was different but similar concept):

```
int &fun(int &z, int y = 18) {
    int x = z;
    x = ++y;
    return z;          // returns reference to z
}

int main() {
    int y = 54;
    int var = ++y; // var = 55, y = 55
    fun(var) = 31; // sets var = 31 (through reference)
    cout << fun(var); // z=31, returns ref to var, prints var
    cout << fun(y)++; // z=55, prints 55, then y++ makes y=56
    cout << var--;    // prints var (31), then var-- makes var=30
}
```

Output: 31 55 31

Step 1: y=54, var = ++y = 55

Step 2: fun(var)=31 → var becomes 31 (reference assignment)

Step 3: fun(var) → z=var=31, x=++y(18)=19, returns z(=var=31) → prints 31

Step 4: fun(y)++ → z=y=55, returns reference to y, prints 55, then y++ → y=56

Step 5: var-- → prints var(=31), then var=30

Note: Student said this was not the exact code — similar concept tha. The key concept being tested: function returning a reference can be used as an lvalue (left side of assignment). Yeh concept samajhna zyada important hai exact answer se.

Hint: Function reference return kare to use assignment ki left side par rakh sakte ho: fun(var) = 31. Yeh sirf reference return mein possible hai, value return mein nahi.

Review 02

Important MCQs — Papers Se Directly

MCQ LIST All MCQs Mentioned in Reviews — Answers Included

Question	Answer
Remove last element of linked list — best case?	O(n) — traverse to end needed
6+4*3/(6-2) postfix answer?	643*62-/+
In BST, if node is greater than root, move to?	Right side
int x[6] — data type of x?	Array of int (int[])
Parameters in function call passed using?	Stack
Compiler uses stack in function call?	True / Correct
Simple case BST — delete leaf node?	Set parent pointer to NULL
Which is NOT a queue operation? (enqueue/dequeue/front/pop)	Pop (it is Stack's operation)
[12,45,17,34,51] remove 45 — which number fills its place?	17 (next node after 45)
Infix 6+4*3/(6-2) — postfix?	643*62-/+

Hint: Yeh MCQs multiple papers mein repeat hue — confirm shuda hain. Inhe raat so jaane se pehle ek baar zaroor padh lena.

Review 03

Review 06

Review 09

LAST MINUTE ADVICE — REVIEWS SE

What to focus on	Why
Stack push/pop output trace	7/9 papers mein tha
Infix to postfix — table format	3+ papers mein exact table format
Reference variable (&) output	Short question almost har paper mein
Queue — NOT pop, it is dequeue	MCQ exact is form mein aaya
BST delete + redraw	Long question 3+ papers mein
MCQs Waqar Siddhu se karo	Review 04 ne specifically kaha
Paradox MCQs (Moaaz file)	Review 09: 12/14 MCQs wahan se aaye